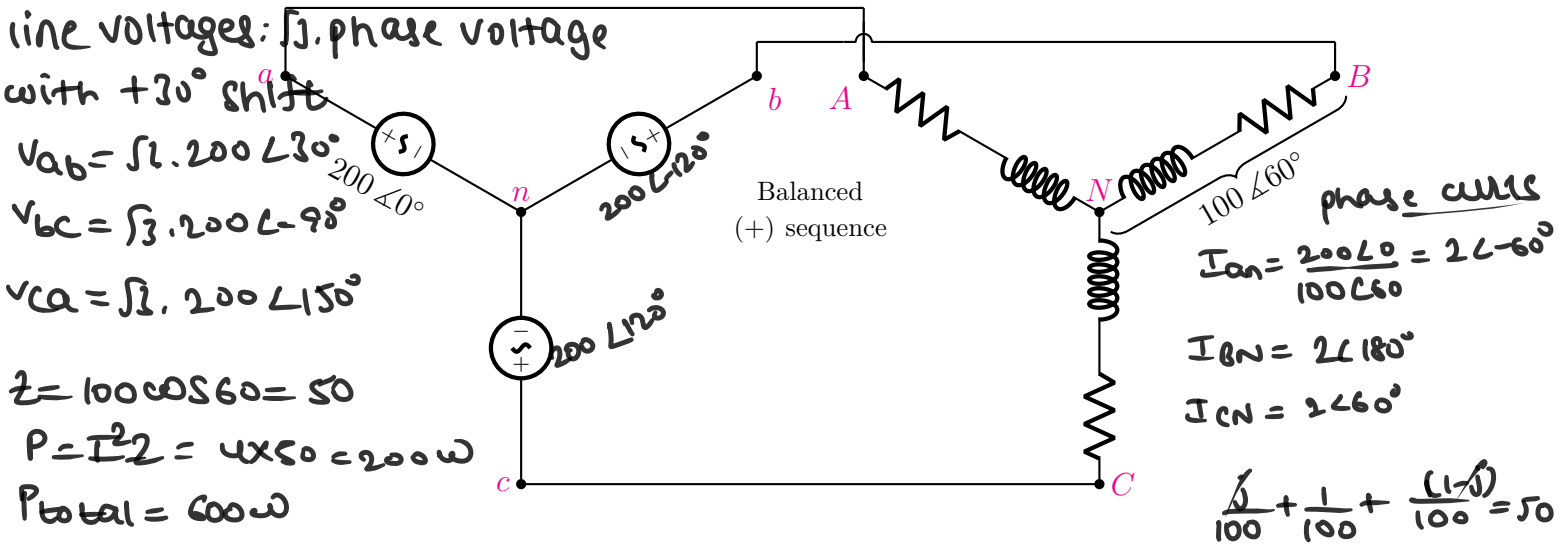
EE2015: Electric Circuits and Networks

Tutorial 10

(October 31, 2025)

1. For the three-phase circuit shown below, find the phase and line currents, and the phase and line voltages. What is the total power dissipated in the three-phase load?



2. A balanced three-phase three-wire system supplies power to a balanced Y-connected load. Each phase contains three loads in parallel, viz., $-j100\,\Omega$, $100\,\Omega$, and $50 + j50\,\Omega$. Assume a positive phase sequence with $\mathbf{V}_{ab} = 400\angle 0°$ V. Find (a) $\mathbf{V}_{an}$, (b) $\mathbf{I}_{aA}$, and (c) total power drawn by the load.

3. Determine the amplitude of the line current in a three-phase system with a 300 V line voltage that supplies 1200 W to a Y-connected load at a lagging PF of 0.8.

4. A balanced three-phase three-wire supply is terminiated with two Δ-connected loads in parallel. Load 1 draws 40 kVA at a lagging PF of 0.8 while Load 2 absorbs 24 kW at a leading PF of 0.9. Let $\mathbf{V}_{ab} = 440\angle 30°$ V. Furthermore, assume that the line resistance is zero. Calculate the following quantities: (a) phase current $\mathbf{I}_{AB}^{(1)}$ for the lagging load, (b) $\mathbf{I}_{AB}^{(2)}$, (c) $\mathbf{I}_{aA}$, and (d) total power drawn by the loads.

5. A balanced three-phase system with a line voltage of 400 V is connected to a balanced Δ-connected load. Assume that the line resistance is negligible. Calculate the line current and per-phase load impedance if the load draws, (a) 1 kW at a lagging PF of 0.85, and (b) 300 W per phase at a leading PF of 0.92.

(a) $1.4^2 \times 500$

(b) $1.406^2 \times 500 = 988.42$

(c) $\mathbf{V}_2 = 400\angle 100°$

$\mathbf{I}_2 = 0.625\angle 178.6°$

$\Rightarrow 0.625^2 \times 500 = 195.31$

(d) $\mathbf{I}_2 = 1.2\angle 45°$

$\Rightarrow 1.2^2 \times 500 = 720$

(e) $V_S = 10I_1 + V_1$

$V_2 = 2I_2$

$I_2 = I_1/5$

$V_2 = 5V_1$

$V_S = 50I_2 + \frac{2I_2}{5}$

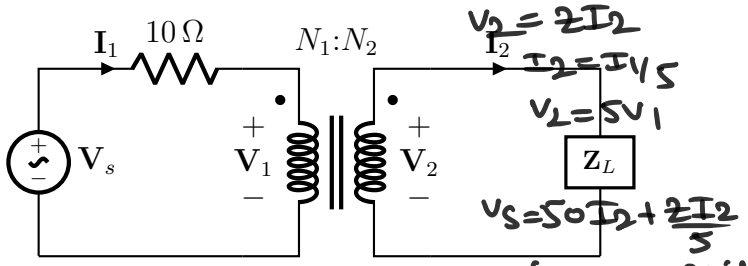
$200 = (50 + 100 - 8j)I_2$

$I_2 = \frac{20}{15 - 8j} = 1.17$

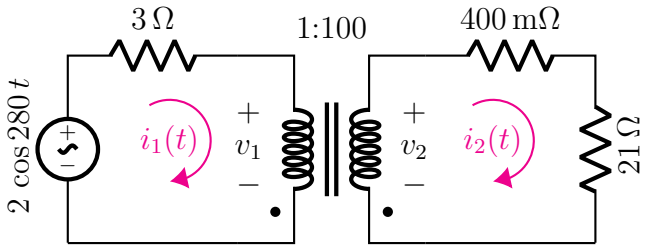
$1.17^2 \times 500 = 691.49$

$640\angle -38.6° \Rightarrow 1.406\angle 78.6°$

6. Let $N_1 = 1000$ turns and $N_2 = 5000$ turns in the ideal transformer shown on the right. If $\mathbf{Z}_L = 500 - j400\,\Omega$, find the average power delivered to $\mathbf{Z}_L$ in each of the following cases: (a) $\mathbf{I}_2 = 1.4\angle 20°$ A rms, (b) $\mathbf{V}_2 = 900\angle 40°$ V rms, (c) $\mathbf{V}_1 = 80\angle 100°$ V rms (d) $\mathbf{I}_1 = 6\angle 45°$ A rms, and (e) $\mathbf{V}_s = 200\angle 0°$ V rms.



7.

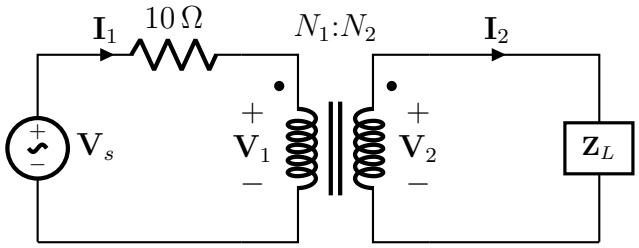


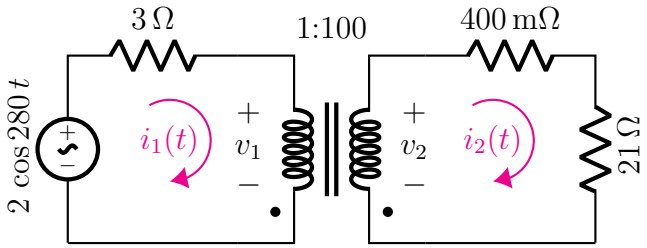
For the circuit given on the left, determine the equivalent circuit in the following two cases: (a) the transformer and secondary circuit are replaced, so that V_1 and I_1 are unchanged, and (b) the transformer and primary circuit are replaced, so that V_2 and I_2 are unchanged.

$V_2 = 100V_1$

$I_2 = I_1/100$

6. Let $N_1 = 1000$ turns and $N_2 = 5000$ turns in the ideal transformer shown on the right. If $\mathbf{Z}_L = 500 - j400\,\Omega$, find the average power delivered to $\mathbf{Z}_L$ in each of the following cases: (a) $\mathbf{I}_2 = 1.4\angle 20°$ A rms, (b) $\mathbf{V}_2 = 900\angle 40°$ V rms, (c) $\mathbf{V}_1 = 80\angle 100°$ V rms (d) $\mathbf{I}_1 = 6\angle 45°$ A rms, and (e) $\mathbf{V}_s = 200\angle 0°$ V rms.



7. 

For the circuit given on the left, determine the equivalent circuit in the following two cases: (a) the transformer and secondary circuit are replaced, so that V_1 and I_1 are unchanged, and (b) the transformer and primary circuit are replaced, so that V_2 and I_2 are unchanged.